

Exponential and Logarithmic Equations



Exponential equations

- 1 Solve $3^{x+1} = 81$ exactly.
 - 2 Solve $5^{2x} = 17$. Give an exact answer and a decimal to three places.
 - 3 Solve $e^{2x} - 5e^x + 6 = 0$ exactly.
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Logarithmic equations

- 4 Solve $\ln(2x - 3) = 1$. Give an exact answer and a decimal to three places.
 - 5 Solve $\log_2 x + \log_2 (x - 6) = 4$, checking for extraneous solutions.
 - 6 Solve $\log(x + 3) - \log(x - 1) = 1$, checking for extraneous solutions.
 - 7 The pH of a solution is $\text{pH} = -\log[\text{H}^+]$. Find the hydrogen-ion concentration $[\text{H}^+]$ of a solution with pH 4.6, in scientific notation to two significant figures.
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More exponential equations

- 8 Solve $2^x = 3^{x-1}$, giving an exact expression and a decimal to three places.
 - 9 Solve $4^x - 3 \cdot 2^x - 4 = 0$ exactly, using the substitution $u = 2^x$.
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Systems and applications

- 10 A bacteria culture grows according to $N(t) = 500e^{kt}$. If the population reaches 2000 after 6 hours, find k (4 d.p.) and hence the time for the population to reach 10,000, to one decimal place.
 - 11 Solve the equation $e^x + e^{-x} = 3$ exactly, using the substitution $u = e^x$ (this expression is related to $2\cosh x$).
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Combined logarithmic and exponential practice

- 12 Solve $\log_3 (x + 2) + \log_3 (x - 2) = 2$, checking for extraneous solutions.

13 Solve $9^x = 27^{x-1}$ exactly, writing both sides with base 3.